MIL-P-21381B(SHIPS)
28 January 1963
SUPERSEDING
MIL-P-21381A(SHIPS)
22 March 1961

MILITARY SPECIFICATION

PLOTTING BOARDS, SHIP'S STATUS AND TACTICAL DISPLAY

(NAVAL SHIPBOARD USE)

1. SCOPE

1.1 This specification covers plotting boards for use in combat information centers or similar spaces in Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-P-5425 - Plastic, Sheet, Acrylic, Heat Resistant.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of.
MIL-S-20708 - Synchros, 60 and 400 Cycle, General Specification.

STANDARDS

MILITARY

MS15571 - Lamp, Incandescent, T-3-1/4, Miniature Bayonet, Single Contact.

DRAWINGS

815-1,853,031 - Plotting and Status Boards, Illumination Control Device for 6-Volt Lamps.
815-1,853,032 - Plotting Board, Ships' Status and Tactical Display.
S2407-756,641 - 36" Edge Lighted Vertical Plotting Board Screen Marking.
S2407-835,550 - Hand Rotatable Own Ship Dial for Vertical Plotting Board Screen Marking.
S2407-1,230,368 - Plotting Surface Design for 60" Vertical Plotting Board.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Applications for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production, a sample shall be tested as specified in 4.3. One of each of the plotting boards, ship's status and tactical display shall be furnished as a sample

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3.2 Description. - Plotting boards, ship's status and tactical display, shall be fabricated in accordance with Drawing 815-1,853,032. Ship's status boards shall be fabricated in any size (rectangular or square in shape) with the height or width of clear marking area from 18 inches to 72 inches. Tactical display plotting boards shall be fabricated in any size (rectangular or square in shape) with the height or width of clear marking area from 24 inches.

3.3 General features. - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein: (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern).

General information
Definitions
Material, general
Substitution of (equal or superior) materials
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Aluminum
Iron and steel
Other metals
Springs (materials)
Plastics
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Dials and pointers
Dial sizes
Washers
Parts-electrical-general
Synchros
Electrical tapes
Dial illumination lamps
Indicator lights and lampholders
Enclosures - general
Enclosure - accessibility
Enclosure mounting
Stiffening grooves
Minimum sheet metal thickness (watertight enclosures)
Through bolting
Ventilation
Size (for submarines)
Threaded devices
Rounded corners and edges
Drilled and tapped holes
Welding
Temperature and humidity
Shock, vibration and inclination (vibration only - nonvital test required)
Personnel protection
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards, connectors and terminals
Synchro connections and markings
Wiring

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Color coding
Dial illumination
Electrical insulation
Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagrams
Assembly drawings
Drawings - manufacturing
Interchangeability and standardization
Manuals
Repair parts
Designation and marking
Workmanship and general examination

3.4 Tactical display plotting board (vertical). -

3.4.1 Vertical plotting boards shall consist of an edge-lighted plotting surface of transparent plastic mounted in an aluminum supporting frame in accordance with Drawing 815-1, 853, 032 The plotting surface shall be inscribed with polar coordinate markings, and shall be suitable for plotting data manually thereon by means of marking with a good grade of grease pencil (China marking type). An own ship's course indicator shall be provided.

3.4.2 Plotting surface. - The plotting surface shall be of a transparent acrylic plastic material in accordance with MIL-P-5425. For boards desired to be fabricated in two halves (see 6.1), the plotting surface (not the supporting frame) shall be composed of two halves arranged for assembly with the joint vertical on board ship. Each half of the plotting surface shall be composed of 3/8-inch thick front sheet and a 1/8-inch thick back sheet. The front and back sheets shall be secured around the outer edges where the plotting surface is fastened in its surrounding frame. At the center of the plotting surface the back sheet shall be secured to the front sheet by means of small screws inserted in holes drilled and tapped in the back side of the front sheet. These screws shall not protrude through the front surface. To provide essential rigidity to the center of the plotting surface and to furnish an additional means of keeping the adjoining inner edges in alignment, the two halves of the plotting surface shall be clamped together at the center point. This clamp shall consist of a bolt, through both front and back sheets, not larger than 1/4-inch in diameter, with a washer on the front and back of the plotting surface. The washers shall not exceed 1 inch in diameter, and the securing nut thereon shall be on the back.

3.4.2.1 Polar coordinate markings. - The polar coordinate markings on the plotting surface shall be applied on the back of the front sheet by means of screen process or similar method satisfactory to the bureau or agency concerned. The polar coordinate markings shall be green in color, and shall consist of range circles and bearing lines in accordance with table I. Bearing lines shall extend from the center point of the plotting surface to, but not beyond, the outer most range circle.

Table I - Polar coordinate markings.

Smaller dimension of board	Diameter of outer-most range ring	Diagram design
24" to but not including 36"	1/ 24"	S2407-756, 641
36" to but not including 48"	1/ 36"	S2407-756, 641
48" to but not including 60"	48"	S2407-1, 230, 368
60" to 72"	60"	S2407-1, 230, 368

1/ Own ship's course indicator (hand-rotatable) when specified for these sizes shall be reduced to a 4-inch diameter on boards that are less than 36 inches on their small dimension.

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3.4.3 Illumination. - The plotting surface shall be edge lighted from top to bottom. Illumination shall be provided by 6.3-volt lamps, spaced 6 inches apart, conforming to type TB-11 (industry No 44) of MS15571. All plotting boards provided under this specification shall contain a red lamp conforming to MS15571-7 in each lamp socket. The mounting of lamp sockets shall conform to Grimes Manufacturing Company part number 20840 or an equivalent which allows removal of individual lamps with their sockets from the back of the board. The resulting position of the lamp shall be such that the lamp filament is in proper alignment with the plastic plotting surface in order to produce the optimum edge-lighting effect The lamps shall protrude through clearance holes provided in the plotting surface near the edges, and within the frame housing, so as to permit removal. (The inherent light transmitting characteristics of an edge-lighted plotting surface is such that it is particularly suitable for application where the preservation of the dark adaptability of the eyes of viewing personnel is of prime importance It is the intent of this specification, with respect to illumination of the plotting surface, that the plotting surface shall appear to have a low level of illumination in a darkened room through the illumination control turned to normal position and, under these conditions only, the printed coordinate design and markings made by the operator with grease pencil will allow light to be transmitted through the plotting surface.) The density of the printed coordinate markings on the plotting surface shall be such that, in addition to being visible and legible at a distance of 8 feet in front of the board, the markings shall also be clearly visible and legible to the operator in the rear of the board when the illumination is adjusted to an intensity between medium and maximum brilliancy. Illumination control shall be provided by means of a variable transformer dimmer control device as shown on Drawing 815-1, 853, 031, which shall be furnished with each board. Wherever feasible, the dimmer control device shall be incorporated in the lower right corner of the frame of the plotting board as shown on Drawing 815-1, 853, 032. Cable access holes with protective grommets shall be provided at both ends of the back of the lower lamp housing, about 1 inch from the sides of the plotting board.

3.4.4 Own ship's course indicator (synchro-driven). -

3.4.4.1 For tactical display plotting boards 48 inches, or larger, an own ship's course indicator shall be provided which shall consist of a synchro-driven, 6-inch diameter transparent plastic dial mounted in the clear space in the upper right-hand area of the plotting surface, as viewed from the front of the board. The dial shall be inscribed with markings and numerals as shown on Drawing S2407-1, 230, 368. The dial shall be countersunk into and flush with the front of the plotting surface.

3.4.4.2 The synchro motor which drives the own ships course dial shall be mounted on the back of the plotting surface. The synchro motor shall be type 23TR(60 or 400 cycle) in accordance with MIL-R-20708 and as specified (see 6.1). Suitable covers shall be provided on all exposed terminals of OSC synchros to prevent inadvertent contact by personnel when equipment is energized.

3.4.5 Own ship's course indicator (hand-rotatable). - When specified, (see 6.1) a hand rotatable own ship's course dial shall be provided in the center of the plotting surface. This dial shall be a 6-inch diameter transparent plastic dial inscribed with markings and numbers as shown on Drawing S2407-835, 550. The dial shall be countersunk into and flush with the front of the plotting surface (see 6.1).

3.4.6 Framing. - A frame shall be provided surrounding the four sides in such a manner that the plotting surface is rigidly secured at the edges. The frame shall function also as an enclosure for the lamps, lamp sockets, and the wiring. The frame shall be fabricated of aluminum to maintain the weight at a minimum, provide the sturdy construction required of equipment destined for shipboard installation, and prevent warping of the plotting surface. Provision shall be made for securing the board in a vertical position by attachment to angles provided by the installing activity.

3.5 Ships' status plotting boards (vertical). -

3.5.1 Ships' status plotting boards shall consist of an edge-lighted plotting surface of transparent plastic material as specified in 3.4.1.

3.5.2 Plotting surface. - The plotting surface and its supporting frame shall be as specified in 3.4.2, except it shall consist of one sheet of plastic material 3/8-inch thick. No markings of any description shall appear on the plotting surface.

3.5.3 Illumination. - Illumination shall be accomplished as specified in 3.4.3.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The inspection procedures for the various tests to be performed on the equipment shall be as specified in 4.6. (The attention of the manufacturer is invited to the applicable paragraphs of MIL-I-983 for information concerning tests such as where conducted, failures, and test conditions)

4.3 Preproduction inspection. - Preproduction inspection shall be conducted on the samples furnished (see 3.1). Examination and tests shall be as follows, performed in the order listed:

Inspection	Requirement paragraph	Test reference
General examination	3.3	MIL-I-983
Accuracy	3.4.2	4.6.2
Operating		4.6.1
Temperature and humidity	3.3	MIL-I-983
Vibration	3.3	MIL-I-983

The results of each examination and test shall be compared with the specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure in the preproduction samples and on future production units.

4.4 Sampling for quality conformance inspection. -

4.4.1 Inspection lot. - All equipment of the same type presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period at the choice of the inspector.

4.4.2 Sampling for group A examination and tests. - A sample of equipments shall be selected from each inspection lot in accordance with table II and subjected to the group A tests listed in 4 5 Lot acceptance shall be based on the requirements of MIL-STD-105.

Table II - Sampling for group A tests AQL (approx.) = 1. 5 percent defective.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group A test)	
		Acceptance number	Rejection number
7 and under	All	-	-
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

The results of each examination and test shall be compared with specification requirements. Failure to conform to the specification requirements for any test shall be counted as a defect and the equipment shall not be offered for delivery. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

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4.4.3 Sampling for group C test. - The group C test will be required by the bureau or agency concerned when the basic design or the material of a vital part has been changed. One complete equipment shall be selected and tested in accordance with 4.5. The results of the test shall be compared with specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.5 Quality conformance inspection. - The sample equipments selected in accordance with 4.4 shall be subjected to the following examination and tests, performed, in general, in the order listed

Inspection	Requirement paragraph	Test reference
Group A		
General examination	3.3	MIL-I-983
Operating test		4.6.1
Group C		
Accuracy	3.4.2	4.6.2
Temperature and humidity	3.3	MIL-I-983

4.6 Test procedures. -

4.6.1 Operating test. - The operating test shall consist of a practical test under normal operating conditions. (The purpose of these tests is to determine that switches, rheostats, illumination adjustments, transmission of light, synchro indicator, and other parts function in accordance with design requirements of this specification.)

4.6.2 Accuracy test. -

4.6.2.1 The polar coordinate markings on the plotting surface and dial markings, where applicable, shall be measured to determine that they are in accordance with 3.4.2, and that such markings are within the limits of accuracy specified therefor.

4.6.2.2 Tests shall be made on the own ship's course indicator on the tactical display plotting board to determine that the rotation is in the proper direction, and that it remain in synchronism with a test synchro transmitter of 5G capacity (or equivalent 400-cycle synchro when required). With the own ship's course indicator energized from the test transmitter, the indicator shall be rotated through 360 degrees at 10-degree intervals, clockwise and counterclockwise, checking the accuracy of rotation against the fixed circumference mark corresponding to zero degrees for each check position. As a result of this test, the mean error shall not exceed 1/2 degree and the maximum error shall not exceed 1 degree.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts -

5.1.1 Plotting boards. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation, and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation, and may conform to the supplier's commercial practice when such meets these requirements.

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5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name, and destination.

5.1.2 Onboard repair parts.- The onboard repair parts shall be preserved and packaged level A, packed level C, and marked in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment requirements (see 6.1).-

(5.2.1 Preservation and packaging, packing, and marking.- The equipment and accessories, repair parts, and technical publications shall be preserved and packaged level A or C, packed level A or B, as specified (see 6.1), and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following

(a) Title, number, and date of this specification.
(b) Type (tactical display or ship's status plotting board) required (see 3.4 and 3.5).
(c) Dimensions of the plotting boards (specify overall and clear marking area, height, and width (see MIL-I-983).
(d) Screening of bearing lines and range circles (see 3.4.2).
(e) Requirement for own ship's indicator, whether synchrodriven or hand-rotatable, and type of synchro motor to be used, if required (see 3.4.4 and 3.4.5).
(f) Quantity of items required.
(g) Quantity of manuals required.
(h) Inventory control point for stock repair parts (see MIL-P-15137).
(i) Whether split construction is required (specify minimum dimensions of access to installation site, if limited).
(j) Whether unit stock number is to be entered on the identification plate (see MIL-I-983).
(k) Preservation and packaging, packing, and marking requirements if other than as required by 5.1 (see 5.2)

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity
Navy - Ships
(Project 6605-N061Sh)

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU

SUBMITTED BY *(Print name and activity)*	DATE

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